

INSPECTION CERTIFICATE EN 10204-3.1 EN 10204:2005-01	Document No.:	465-1360646/001/P	Inspection No.:	Page: 1 / 3
	Purchase Order No.:	6591	Manufacturer:	
	Benteler Order No.:	48002775	Manufacturer's brand:	
	Dispatch Note No.:		Stamp of the inspection representative:	
	Product:	SEAMLESS STEEL TUBES	Steelmaking process:	

Terms of delivery:	ASTM-A 53-2018, ASME SA-53, ASME BPVC.II.A-2021, ASTM-A 106-19a, ASME SA-106, ASME BPVC.II.A-2021, API Specification 5L Forty-Sixth Edition April 2018 incl. Errata 1, May 2018, PSL 2, ANSI/NACE MR0175/ISO 15156-1: 2015, ANSI/NACE MR0175/ISO 15156-2: 2015, ANSI/NACE MR0103/ISO 17945:2015
Dimensions-tolerances:	ASTM-A 106-19a, ASME SA-106, ASME BPVC.II.A-2021, ASTM-A 53-2018, ASME SA-53, ASME BPVC.II.A-2021, API Specification 5L Forty-Sixth Edition April 2018 incl. Errata 1, May 2018
Steel grade:	GRADE B, GRADE C, GRADE BN, GRADE X42N
Delivery condition:	N
Product marking:	FS: BENTELER BENTELER-Sign API 5L-1072 API-Sign Month/Year Conditions Dimension: O.D. * t inch Steel Grade PSL2 SMLS Heat No. Tube length feet Test Pressure PSI Production No. Customer's material no.

AEZ = Etching ink marking FK = color marking FS = paint stenciling FSD = Color jet printer FSI = Color jet printer inside LK = Laser marking PKE = tag marking PS = die stamp TS = Ink jet spray marking

Item	Dimensions	Heat No.	Test pressure	
			PSI	sec
0002	2 NPS - SCH 80 35.000 FT - 45.000 FT	314355	2970	5

Country of melt and pour

Melted & Manufactured in USA
Nucor, Jewett, TX, US

Heat analysis [%]

Item	Heat No.	C	SI	MN	P	S	CR	MO	NI	CU	V	NB	TI	B
0002	314355	0.170	0.254	0.97	0.011	0.002	0.12	0.03	0.08	0.18	0.004	0.002	0.002	0.0003

1. Formula: $CE\ IIW = C + (Mn/6) + ((Cr + Mo + V)/5) + ((Cu + Ni)/15) \leq 0,43\ \%$
2. Formula: $Cr + Cu + Mo + Ni + V \leq 1.00\ \%$
3. Formula: $Nb + V \leq 0.06\ \%$
4. Formula: $Nb + Ti + V \leq 0.15\ \%$

Formula results

Item	Heat No.	1. Formula	2. Formula	3. Formula	4. Formula
0002	314355	0.380	0.414	0.006	0.008

Product analysis [%]

Item	Heat No.	C	SI	MN	P	S	CR	MO	NI	CU	V	NB	TI	B
0002	314355	0.180	0.270	0.98	0.010	0.002	0.12	0.02	0.08	0.18	0.004	0.001	0.003	0.0002
0002	314355	0.180	0.280	1.00	0.015	0.003	0.14	0.02	0.08	0.19	0.004	0.002	0.003	0.0003

1. Formula: $CE_{IIW} = C + (Mn/6) + ((Cr+Mo+V)/5) + ((Cu+Ni)/15) \leq 0.43 \%$
2. Formula: $Cr + Cu + Mo + Ni + V \leq 1.00 \%$
3. Formula: $Nb + V \leq 0.06 \%$
4. Formula: $Nb + Ti + V \leq 0.15 \%$

Formula results

Item	Heat No.	1. Formula	2. Formula	3. Formula	4. Formula
0002	314355	0.389	0.404	0.005	0.008
0002	314355	0.397	0.434	0.006	0.009

Test results

The tubes have been submitted to a leak tightness test by:	Hydrostatic test: acc. to API 5L, Test pressure/Time-record, the test records/charts shall be available for examination by the inspector			PASSED
The tubes are non destructive tested:	ET - Test standard: acc. to API 5L; ET - Test method: acc. to ASTM-E 309; ET - Test drilled hole: 0.125 IN.; ET - Notch shape/hole: drilled hole			PASSED
Visual inspection:	PASSED	Bending test:	PASSED	Dimensions examination: PASSED

Results of mechanical testing

Tensile test longitudinal at RT, Strip test specimen

Item	Specimen No.	Heat No.	Specimen dimensions	Yield strength	Tensile strength	Elongation	1. Formula	Rt 0,5/Rm
Requirements			IN	RT 0,5 % PSI 42100-71800	Rm PSI 70000-95000	A2" %		MAX 0.93
0002	000011	314355	0.749 x 0.217	52504	77885	32.00	19.00	0.67

Elongation formula

1. Formula $AF = 625000 * AXC^{0,2} / U^{0,9}$

Protection

outside surface varnished

Item	Heat No.	Production No.
0002	314355	15026241

Remark

NACE Standard: Hardness acc. to ANSI/NACE MR0175/ISO 15156: HRC max 22 as per ASTM A370 table 2&3., Hardness acc. to ANSI/NACE MR0103/ISO 17945: HRC max. 22 as per ASTM A370 table 2&3.; Certificate remarks: material free from mercury contamination, Data converted from SI units, No weld repair has been carried out;
Certificate-Remark: LONGHORN